

**R User Group at
Harvard Data Science Initiative
presents**

AN INTRO TO

WITH
DR. BARRET
SCHLOERKE



Outline

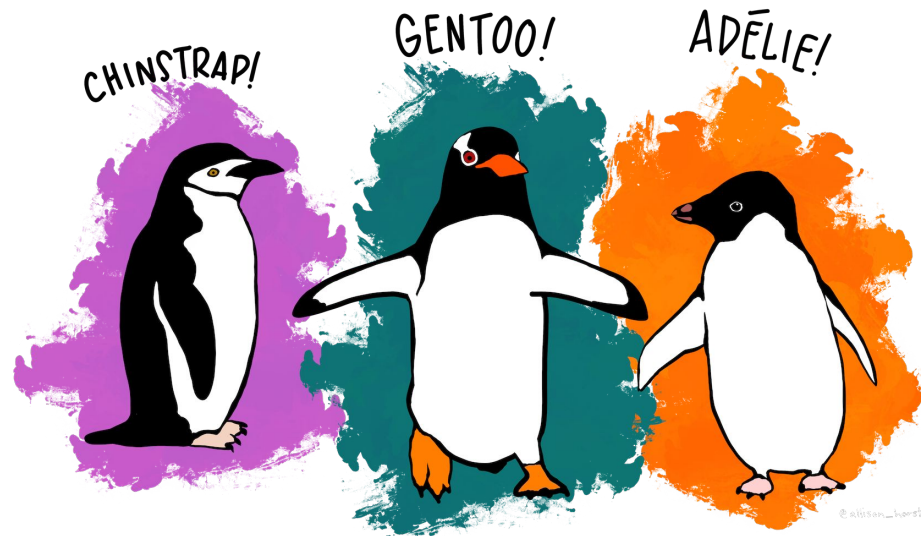
- Set the stage
- What is an API?
- Introduction to APIs with Plumber
- New Plumber features
- API Deployment
- Additional Resources



Photo by [Samuel Sianipar](#) on [Unsplash](#)

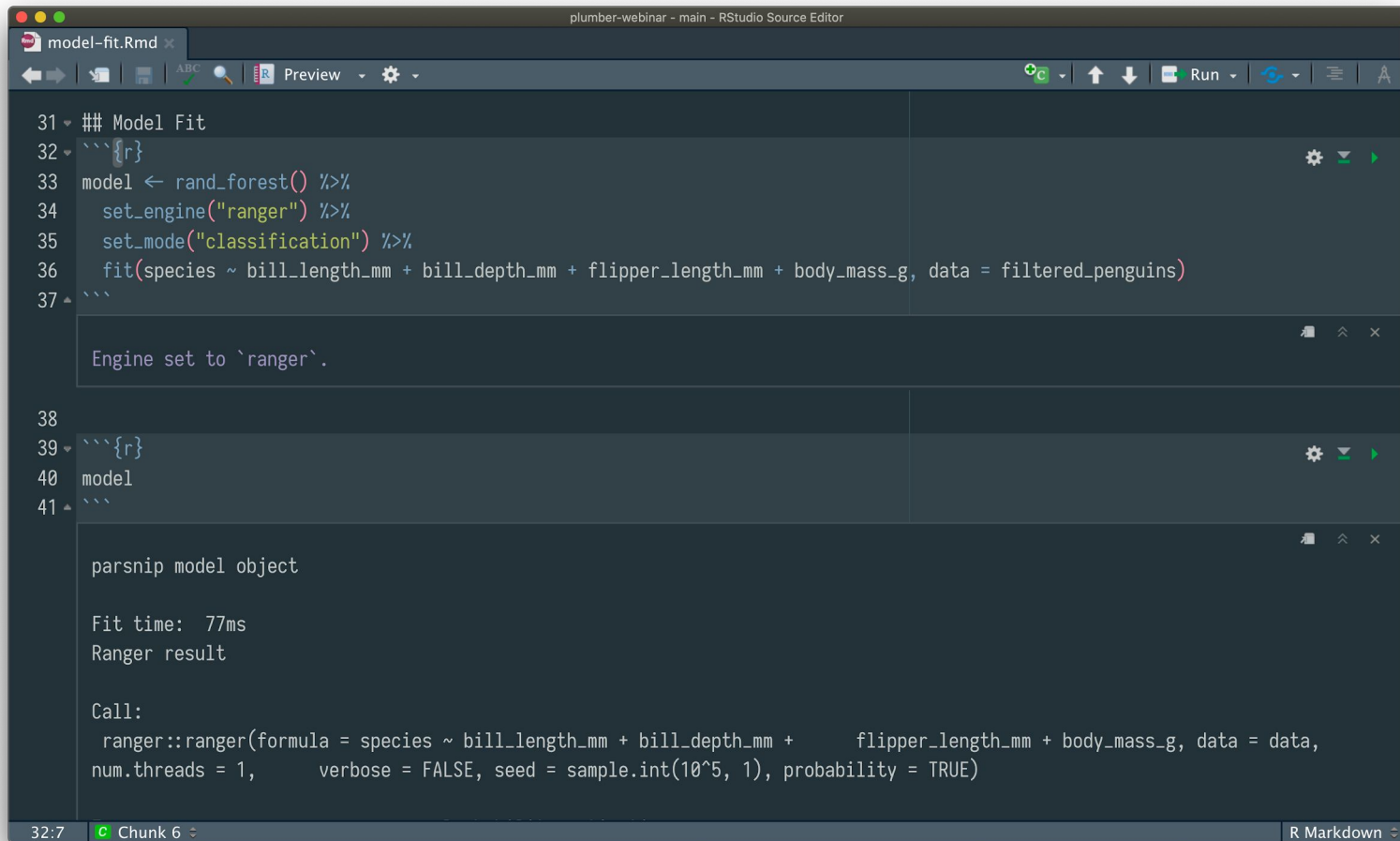
Meet the Palmer Penguins

- [allisonhorst/palmerpenguins](https://allisonhorst.github.io/palmerpenguins/)
- 344 observations for 3 different penguin species
- Simple classification model using a random forest



species	island	bill_length_mm	bill_depth_mm	flipper_length_mm	body_mass_g	sex	year
Adelie	Torgersen	39.1	18.7	181	3750	male	2007
Adelie	Torgersen	39.5	17.4	186	3800	female	2007
Adelie	Torgersen	40.3	18.0	195	3250	female	2007
Adelie	Torgersen	NA	NA	NA	NA	NA	2007
Adelie	Torgersen	36.7	19.3	193	3450	female	2007
Adelie	Torgersen	39.3	20.6	190	3650	male	2007
Adelie	Torgersen	38.9	17.8	181	3625	female	2007
Adelie	Torgersen	39.2	19.6	195	4675	male	2007
Adelie	Torgersen	34.1	18.1	193	3475	NA	2007
Adelie	Torgersen	42.0	20.2	190	4250	NA	2007

Do the data science (build the model)



The screenshot shows the RStudio Source Editor with a file named 'model-fit.Rmd'. The code is written in R and uses the `ranger` package for a random forest model. The code is as follows:

```
31 ▾ ## Model Fit
32 ▾ ```{r}
33 model <- rand_forest() %>%
34   set_engine("ranger") %>%
35   set_mode("classification") %>%
36   fit(species ~ bill_length_mm + bill_depth_mm + flipper_length_mm + body_mass_g, data = filtered_penguins)
37 ▴ ```
```

Below the code, the output of the first chunk is displayed:

```
Engine set to `ranger`.
```

The second chunk of code is partially visible:

```
38
39 ▾ ```{r}
40 model
41 ▴ ```
```

Below the second chunk, the output is displayed:

```
parsnip model object

Fit time: 77ms
Ranger result

Call:
ranger::ranger(formula = species ~ bill_length_mm + bill_depth_mm + flipper_length_mm + body_mass_g, data = data,
num.threads = 1, verbose = FALSE, seed = sample.int(10^5, 1), probability = TRUE)
```

The status bar at the bottom indicates '32:7' and 'Chunk 6'.

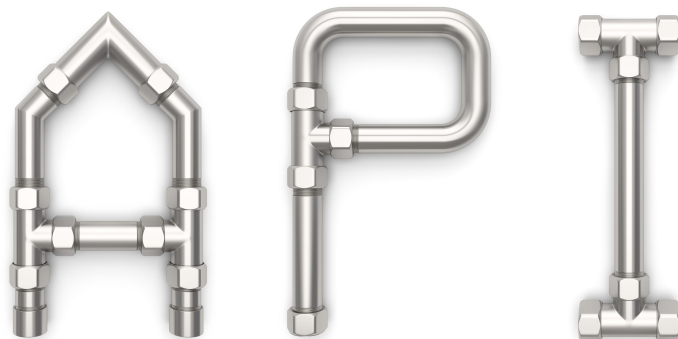
Wait.
Now what?



Photo by [tian kuan](#) on [Unsplash](#)

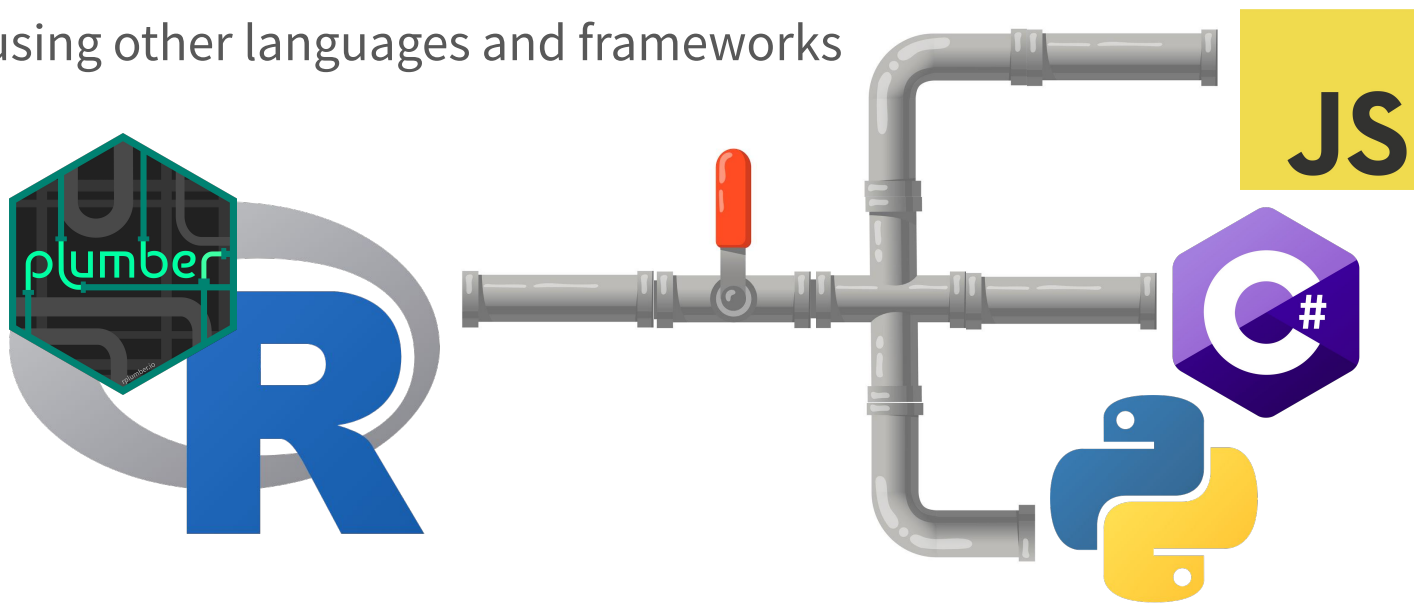
What is an API?

- **A**pplication **P**rogramming **I**nterface
- Web APIs over HTTP are a standardized way for different computers to communicate
- A client sends a request, a server sends a response
- Web browsers make API calls to every website you visit



Why do I care?

- APIs allow your work to be used by a wide range of tools and technologies, not just R
- Dramatically reduces the handoff between work done in R and tools built using other languages and frameworks



Introduction to APIs with Plumber

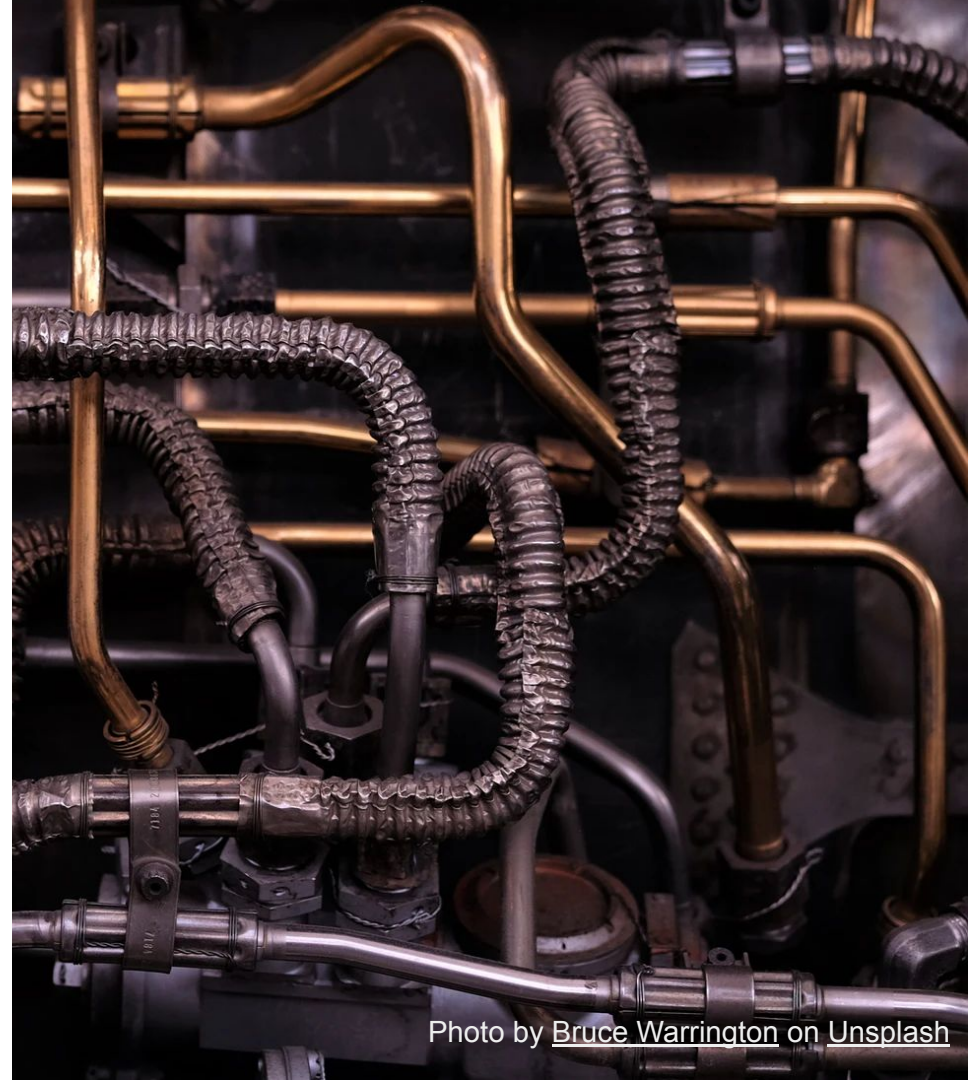
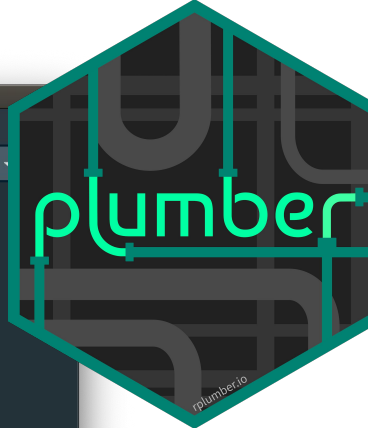


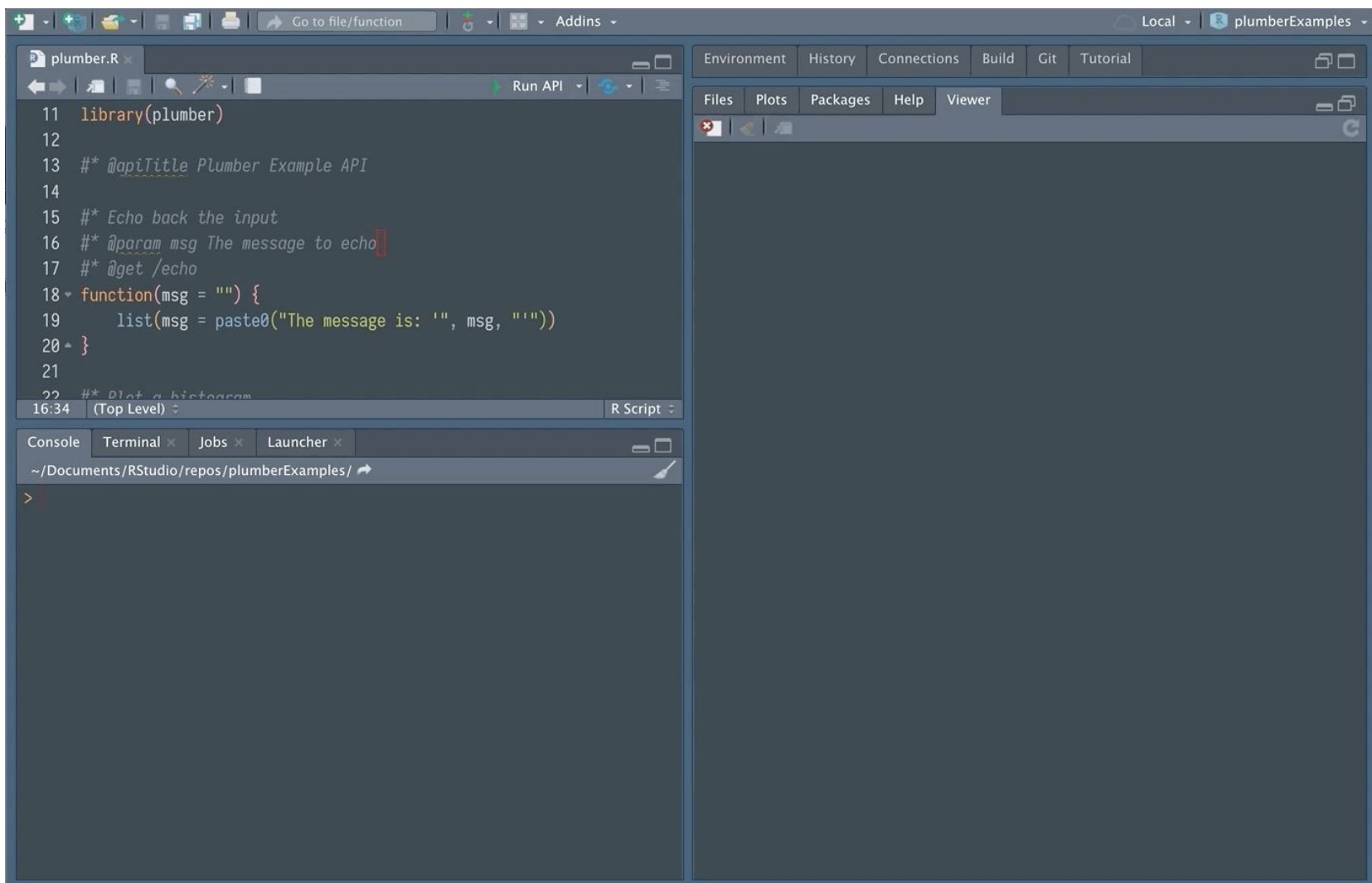
Photo by [Bruce Warrington](#) on [Unsplash](#)

```
plumber.R x
plumberExamples - master - RStudio Source Editor
Run API

11 library(plumber)
12
13 /* @apiTitle Plumber Example API
14
15 /* Echo back the input
16 /* @param msg The message to echo
17 /* @get /echo
18 function(msg = "") {
19   list(msg = paste0("The message is: ", msg, ""))
20 }
21
22 /* Plot a histogram
23 /* @serializer png
24 /* @get /plot
25 function() {
26   rand <- rnorm(100)
27   hist(rand)
28 }
29
30 /* Return the sum of two numbers
31 /* @param a The first number to add
32 /* @param b The second number to add
33 /* @get /sum
34 function(a, b) {
35   a + b
36 }
37
38 # End of file
```



- Write standard R functions
- Use special comments (*/**) and tags (*@apiTitle*)
- Plumb it!



Penguin API

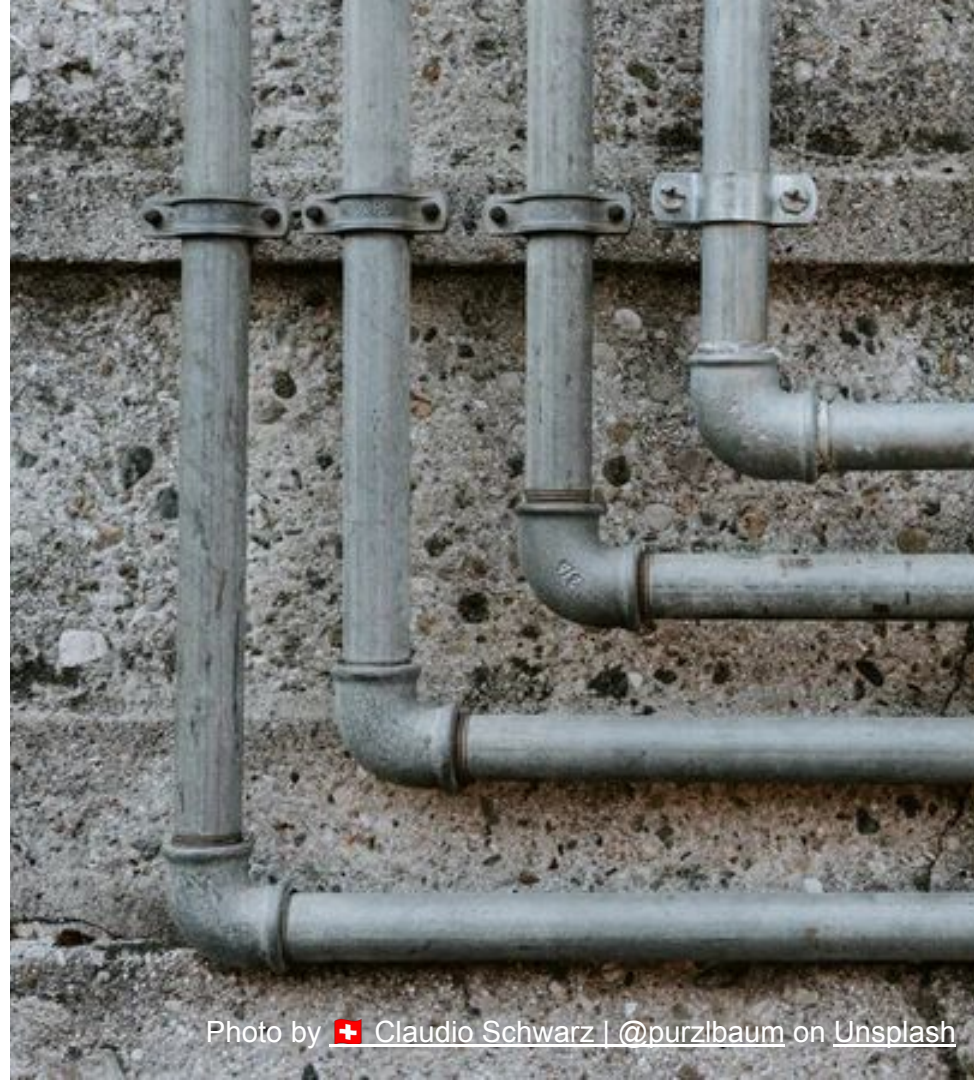


Photo by  Claudio Schwarz | @purzlbaum on [Unsplash](#)

Deployment



Photo by [Dieter K](#) on [Unsplash](#)

Deployment options

- [RStudio Connect](#)
- [Docker](#)
- [DigitalOcean](#)

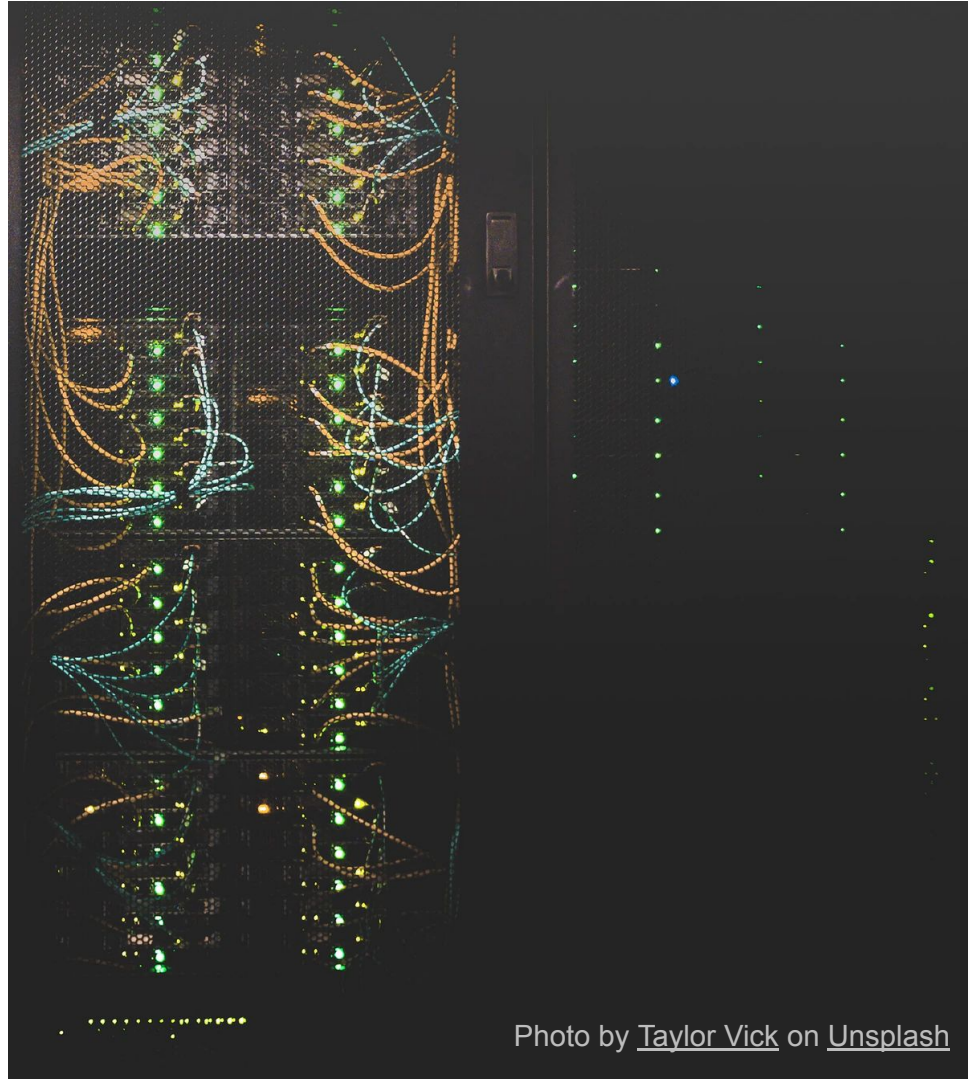


Photo by [Taylor Vick](#) on [Unsplash](#)

RStudio Connect

- Push button deployment
- Dependency management
- Scalability
- Git integration
- Authentication and permission controls
- Publish and share RMarkdown, Shiny, Plumber, Jupyter notebooks, Flask, Dash, Streamlit
- rstudio.com/products/connect

The screenshot displays the RStudio Connect web interface for an API named 'predict-penguins'. The top navigation bar shows 'Content / predict-penguins' and a user profile 'JB'. Below this, a dark header contains the URL 'https://colorado.rstudio.com/rs' and a search bar. The main content area is titled 'Penguin Predictions 1.0.0' and includes an 'API Description' section with a 'health-check' endpoint (GET /health-check) and a 'predict' endpoint (POST /predict). The right sidebar contains 'Sharing settings' (set to 'Anyone - no login required'), a search bar, and sections for 'Who can view or change this API' (showing user 'JB James Blair james') and 'Who runs this content on the server' (set to 'The default user rstudio-connect'). At the bottom, the 'Content URL' is shown as '/predict-penguins/' with a full URL 'https://colorado.rstudio.com/rsc/predict-pe...' and a 'Copy' button.

Resources



Photo by [Christophe Dion](#) on [Unsplash](#)

Additional Resources

- <https://www.rplumber.io/>
- <https://github.com/rstudio/plumber>
- <https://github.com/sol-eng/plumberExamples>
- <https://github.com/meztez/plumberDeploy>
- <https://github.com/meztez/rapidoc>
- <https://github.com/rstudio/webinars>
- <https://community.rstudio.com/tag/plumber>
- <https://rstudio.com/conference/>



```
#' @apiTitle Plumber Beginner
```

```
## @get /ask
```

```
function(question) {  
  barret::answer(  
    question  
  )  
}
```

